

Design Document

Pirates: Sea Rail

This document entails the game design and technical implementations of Pirates: Sea Rail.

Overview

At its highest concept, Pirates: Sea Rail is an inspiration of Turn-Based Gacha RPGs, predominantly, Honkai Star Rail.

The basic exercise we want to express via this project is integrated backend, specifically those used in Gacha systems and how they can track player resources and information across time; merging a live server running with an actual game in of itself.

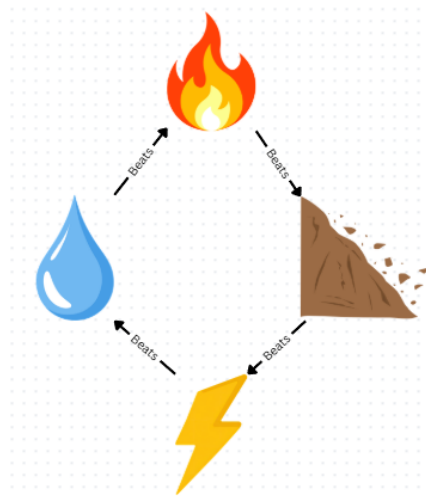
Gameplay-wise, you are a stranded *Outsider* who woke up in a neighboring country. After receiving help from the locals and repaying your debt by staving off pirates, you realized that there is more to this than meets the eye. A whirlwind of **Sea Curses**, **Shadow Empires**, and **Piracy** sails the world.

Combat

Pirates: Sea Rail takes the approach of Honkai Star Rail in doing its battle system.

A character enters the battle with a maximum of three skills on the right á la Chrono Ark, a party facecard on the left, with each character's portrait being in front of an Energy Circle. Each circle is colored differently and have different symbols either. Menu navigation is done via mouse, and ultimates can be enacted via 1, 2, 3, ...

Then, we get to the main draw of the battle system: the elemental interactions. There are only 4 main elements: Water, Earth, Lightning, Fire. The main principle is the following rectangle:



This reveals an enemy's hierarchy/weaknesses. Each enemy is assigned at least one Elemental Signature of a combination of the elements above. And there is a special element: **Wind**, which interacts differently to the other four.

Upon hitting an enemy enough times to "Break" them with their respective weakness, enemies will enter a "Signature Loss" state, where you can inflict an "Elemental Mark" after their first *break* instance*. When two elemental marks, combine they can apply one of the following status effects on said enemies:

First \ Second	Fire	Water	Earth	Lightning
Fire	-	Douse*	Smothered	Overload
Water	Evaporate	-	Muddied	Electrocute*
Earth	Burn*	Damp	-	-
Lightning	Overload	Electrocute	Isolate*	-

These are the details of the status effects. All of them do not stack (so you can't do Burn twice):

1. Evaporate/Douse: Amplifies the inflicting skill's DMG by 1.5x.
2. Burn: Creates a DoT which deals 20 Base Power (of the Source) DMG for 5 turns.
3. Smothered: Decreases user's ATK by 50% for one turn.
4. Overload: Creates an explosion which deals 1.2x the inflicting skill's DMG on an AoE.
5. Damp: Decreases user's DEF by 33% for one turn.
6. Muddied: Decreases user's SPE by 33% for one turn.
7. Electrocute: Amplifies the inflicting skill's DMG by 1.3x, then creates a DoT which deals 10 Base Power for 3 turns.
8. Isolate: Stuns the user for one more turn.



The Damage

Enemies will take 1.2x damage to attacks that are of their weakness, with 1.0x damage to attacks that are not of their weakness, with 0.8x damage to attacks that are resistant to their weakness. This is, at default, calculated by the element that they have. The elements that they wield holds the value of their resistance towards attacking elements. These are the *default rates*, and there could be a RES stat which either heightens that resistance or lowers it.

The Break

Enemies can only be hit with the element of their weaknesses. This means an Earth-element enemy can only be broken with a Fire element. The skill which *breaks* the enemy will always inflict the states marked by (*) according to the skill's element. Against an Earth/Lightning enemy (which has a weakness of both Earth and Fire), if the breaking skill is an Earth element, it will always inflict *Isolate*; if the breaking skill is a Fire element, it will always inflict *Burn*.

Signature Loss

Once enemies are broken--and after they receive their break--they are now in a state of Signature Loss for one turn. In this phase, all elemental attacks applied to the enemy will inflict an Elemental Mark. Upon two Marks meeting, an Elemental Reaction will occur. Some characters have the ability to manipulate these status effects to greater heights.

Reaction

The Reaction is the total DMG that occurs due to the meeting of two Elemental Marks. Elemental Marks stores the information of their applicator, and uses it to calculate DMG scaling off the MAG stats. An elemental reaction DMG is not owned by anyone and therefore *cannot* CRIT. All reaction DMG use the same calculation,

Wind Element

The **Wind** element has the innate ability of camouflaging into another element. In this case, the Wind element is able to apply another enemy's Elemental Signature as an Elemental Mark. Take two enemies: one Fire, one Earth. Using a Wind attack will change the element into that of the target. If an AoE wind attack hits the Fire enemy, it will deal Fire damage to the Earth enemy and, if broken, apply Fire Mark to that Earth enemy. If an target already has a mark, the Wind attack will re-apply that mark to all other targets.

Stats

Each character has these stats:

- ATK determines the DMG Output of Base Power.
- DEF mitigates incoming DMG.
- MAG determines the DMG calculation of Reactions.
- SPE determines the turn order in battle.
- PEN determines the DEF penetration.
- CRIT Rate. determines the chance for critical hit
- CRIT DMG. determines critical damage
- Weakness Bar (Enemy Only)
- RES (Enemy Only)
- Break Efficiency (Hidden)
- Incoming Healing (Hidden)
- Shield

Mana System

Mana is the team wide resources/systems. The core idea and philosophy of a Mana/Cost system is that they're supposed to limit.